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ORIGINAL RESEARCH ARTICLE

Finite element modeling and validation of a four-bar linkage prosthetic knee under static and cyclic strength tests

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This work presents a procedure to simulate the static and cyclic strength tests implemented to ensure the structural integrity of a four-bar linkage prosthetic knee. The test protocols used in this work follow a standard defined by the International Organization for Standardization (ISO). This standard is titled 'Prosthetics—Structural Testing of Lower-Limb Prostheses—Requirements and Test Methods' or shortly called ISO 10328:2006. The purpose of this standard is to guarantee that a prosthesis is able to survive a random severe load and has a sufficient fatigue life. Finite element method, which is a numerical technique used to model a physical system, is employed as a primary computational tool to simulate the prosthesis under the tests. The method of explicit nonlinear transient stress analysis is applied to determine the strength of prosthesis. Consequently, the finite element model reveals the stress distributions induced on the prosthesis and predicts its fatigue life. Besides, to examine the accuracy of the model, the simulation results, particularly structural strains, are validated with those obtained from experiments. In each testing condition, five repetitions are performed to ensure the result consistency. The validation results confirm the fidelity of the proposed finite element model. The average of absolute percentage errors between the simulation and experimental results in all testing scenarios is estimated to be 22%.

Keywords: *prosthetic knee; artificial limb; ISO 10328:2006; finite element analysis; static strength test; cyclic strength test*

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To improve the life quality of an above-knee amputee, it is widely accepted that a prosthetic knee is one of the most essential assistive technologies to allow him/her to regain mobility. In addition to the stability, comfort, appearance, and ease of use of the prosthesis, the safety is another important design consideration. To ensure its safety, the International Organization for Standardization (ISO) has defined a standard called 'ISO 10328:2006 Prosthetics—Structural Testing of Lower-Limb Prostheses—Requirements and Test Methods' (1). This particular standard outlines a procedure to physically conduct the static and cyclic strength tests. With the aid of Finite Element Method (FEM), these tests can further be implemented in a virtual environment.

In a typical cyclic test, at least 35 days are required to complete such tests based on the loading frequency of 1 Hz. The testing time can be significantly reduced with the use of FEM, accelerating the development time and saving some costs.

After World War II, tremendous efforts were spent on the design and construction of more functional prostheses in response to the demands of veterans (2). As a consequence, the design and function of a prosthetic knee was greatly improved. Besides, after the space race era, a numerical technique known as FEM (3) has been substantially generalized for the physical system modeling in a broad range of physics and engineering disciplines, for instance, structural analysis, fluid dynamics, heat

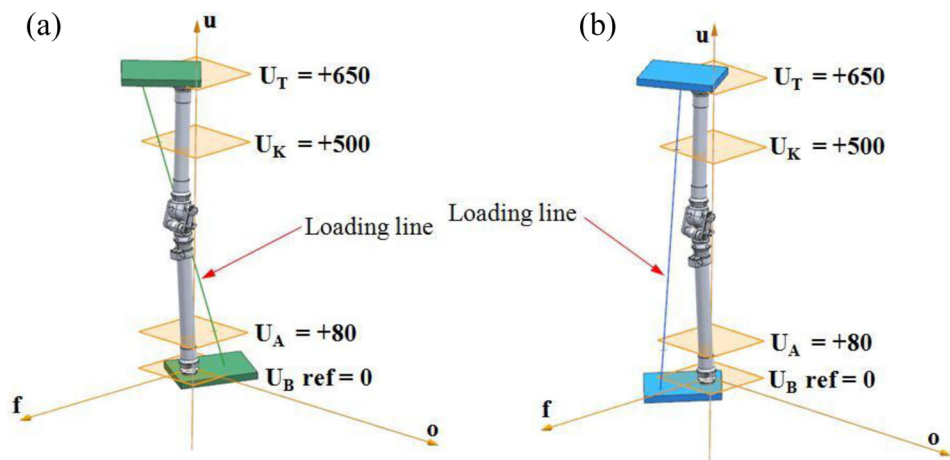


Figure 1. Prosthetic knee with defined reference planes and loading line. (a) Condition I. (b) Condition II.

transfer, etc. The field of biomedical engineering has also adopted this powerful computational tool that allows biomedical engineers to analyze and understand various relevant phenomena. Further, FEM plays a vital role in the process of prosthetic design and development. FEM is applied to analyze stress distributions on a prosthetic knee implant or knee joint replacement (4–8). Better understanding of the stress contour leads to the development of more resilient knee implants.

The design of a prosthetic socket is also a major consideration, since it is the primary interface between an amputated limb and a prosthesis. Finite element modeling of such an interface is constructed to determine pressures and shear stresses on a stump (9–12). An optimal limb-socket load transfer comforts an amputee during use. As one may see, the application of FEM is rather well-suited in the prosthetic design. However, its use and integrity in the design of a prosthetic knee for an above-knee amputee is still lacking. Hence, this paper gives details for establishing the virtual static and cyclic strength tests of the prosthetic knee. Finite element modeling is used to simulate a four-bar linkage prosthetic knee under the tests. Moreover, the results of experimental validation of this specific application are presented to verify the fidelity of the model.

Materials and methods

Static and cyclic strength tests

In this work, a guideline to conduct the static and cyclic strength tests on a prosthetic knee is adopted from the ISO 10328:2006 standard. Basically, the purpose of the static

test is to guarantee that a prosthesis is able to sustain a random severe load that it may experience in its life, and its fatigue strength is proofed in the cyclic test. Referring to the standard, a single test force applied through a pre-defined loading line is required to create compound loadings on the prosthesis. However, the magnitude and direction of the test force are varied, depending upon test conditions and load levels. For a complete structure test, two test conditions are necessary. These tests represent the peak loads exerting on the prosthesis at the beginning and end of the stance phase of walking, hereafter called condition I and condition II, respectively. In this work, the



Figure 2. Model of a prosthetic knee. (a) CAD model. (b) Finite element model.

Table 1. P4 loading level for static and cyclic strength test

Loading condition	Static test force (N)	Cyclic test force (N)
I	2,065	1,230
II	1,811	1,085

Table 2. Mechanical properties of materials

	Stainless steel	Plastic	Aluminum
Modulus of elasticity (GPa)	200	110	70
Poisson ratio	0.30	0.42	0.33
Density (kg/m ³)	7,850	950	2,800

P4 load level is selected to limit the maximum weight of an amputee to 80 kg.

To define the loading line of each test condition, an instantaneous center of rotation (13, 14) must be first located in the case of a four-bar linkage prosthetic knee. Four planes including a reference plane are subsequently defined according to the standard, and one of the planes must pass through the instantaneous center. A point is further assigned on each plane. By connecting these points, the loading line is specified. Figure 1 illustrates an example of the four-bar linkage prosthetic knee with the defined reference planes and loading line.

Furthermore, Table 1 summarizes the magnitude of test force of the P4 load level for each test condition in the static and cyclic strength tests. It is important to note that the test force in the static test must be slowly applied to avoid the dynamic excitation of the prosthesis. In the cyclic test, the prosthesis is subjected to 3×10^6 cycles of mechanical endurance testing to determine its fatigue strength. In addition, the prosthesis must maintain the corresponding proof test force for 30 ± 3 seconds after the cyclic test is completed. To pass the standard, a prosthetic knee is needed to withstand both of the static and fatigue tests with the loading conditions I and II without any failures of any parts. Besides, the total deformation between both ends must not exceed 5 mm.

Finite element modeling

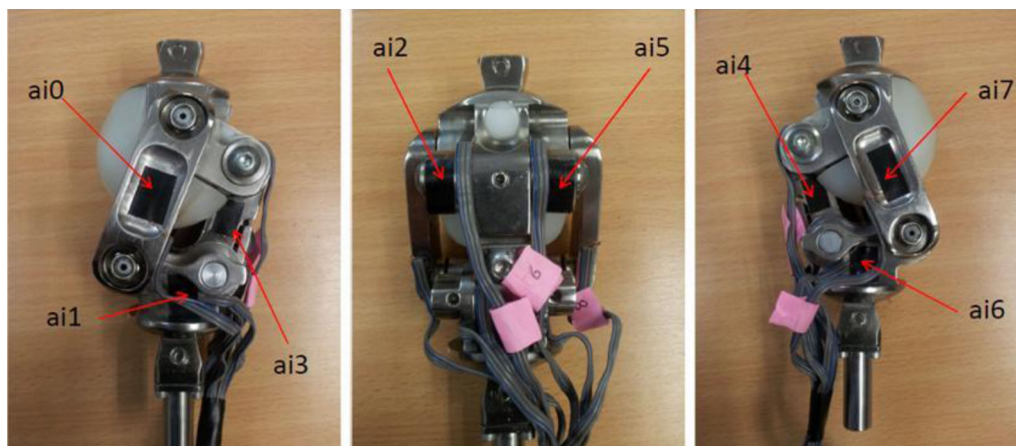
A three-dimensional solid model of a four-bar linkage prosthetic knee is first formed as shown in Figure 2a. The configuration of this solid model must be complied with

the ISO 10328:2006 standard briefly introduced in the previous section. Illustrated in Figure 2b, the finite element model is further developed based on the solid model in a Computer-Aided Engineering (CAE) software package called LS-DYNA. This finite element model is meshed with three-dimensional tetrahedron elements. Inherently, this type of element consists of four nodes per element, and each node has three degrees of freedom (3). The model is entirely composed of 144,799 nodes and 604,250 elements.

Furthermore, to obtain an accurate prediction of the stress distribution on the prosthesis, proper boundary conditions are essential. Two lever arms are imposed on the prosthesis model to ease the simulations of strength tests as shown in Figure 1. A test force is applied in the direction of the loading line defined earlier on the top plate. Translational constraints are given to the model on the bottom plate, limiting its translational movement in any direction. Besides, mechanical properties of the prosthesis are necessary to complete the modeling. Table 2 summarizes the mechanical properties of materials utilized in the fabrication of the prosthesis. The simulations in this work employ the method of explicit nonlinear transient stress analysis to predict the strength of the prosthesis, since this method consumes less computational resources and is able to deal with a nonlinear contact problem. Furthermore, a proper mass scaling is implemented to reduce the dynamic effects of this method (15). In the cyclic test, the maximum stress is first obtained from this analysis in LS-DYNA, and is then given to the MSC Fatigue module to estimate the fatigue life of the prosthesis. The S–N curve of stainless steel is estimated based on an empirical formula provided in (16).

Experimental setup

To verify the fidelity of strength prediction of the finite element model introduced in the previous section, an experimental validation is required. In this work, structural

**Figure 3.** Prosthetic knee instrumented with strain gauges.

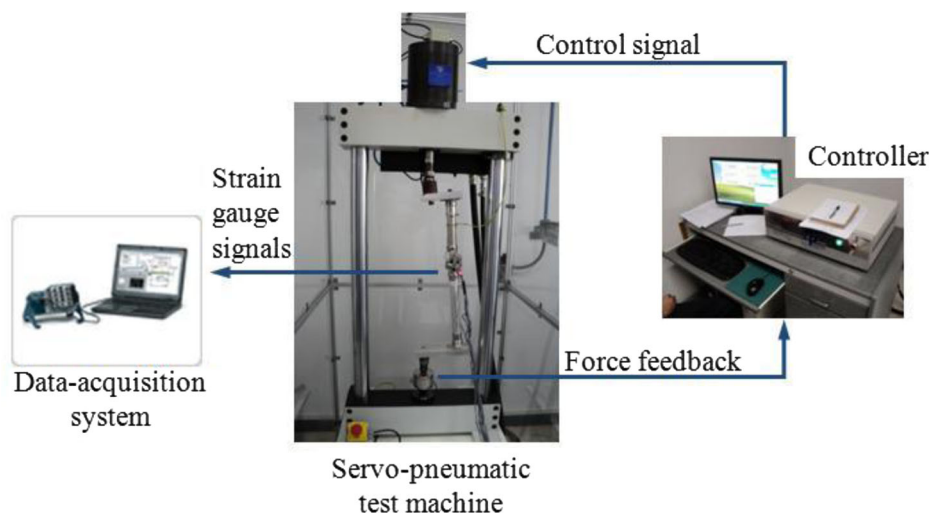


Figure 4. Experimental setup.

strains are used as a metric to serve this validation purpose. Eight three-wired thin-foil strain gauges are positioned on a four-bar linkage prosthetic knee as shown in Figure 3 along with their identification numbers (ai0, ai1, ai2, . . . , ai7). All of the strain gauges are further connected to a National Instruments' NI 9235 module to measure the strains. The strain data are then recorded on a computer through the LabVIEW SignalExpress software at a sampling rate of 100 Hz.

To physically conduct the strength tests of the prosthetic knee, a bench-top servo-pneumatic test machine, which is illustrated in Figure 4, is used. This machine is manufactured by Si-Plan Electronics Research Ltd. and is capable of managing both static and cyclic tests. An arrangement of the instrumented prosthetic knee on the test machine is shown in Figure 4. The test procedures are applied in accordance with the ISO 10328:2006 standard

described in 'Materials and methods' section. Besides, to ensure the repeatability of the experimental results, each test condition is repeated five times.

Results

Simulation results

The simulation results of the static test are first presented. Figure 5 illustrates the stress distributions on the prosthesis subjected to the loading conditions I and II. The stress in the figures is the von Mises type. The maximum stresses are 153 MPa in both conditions. One can see that the maximum stresses are still below the yield strength of stainless steel that is a main material used to manufacture this prosthetic knee. Its yield strength is indicated in Table 2. Concluded from these results, this prosthesis sample is predicted to be sufficiently strong to survive the static strength test.

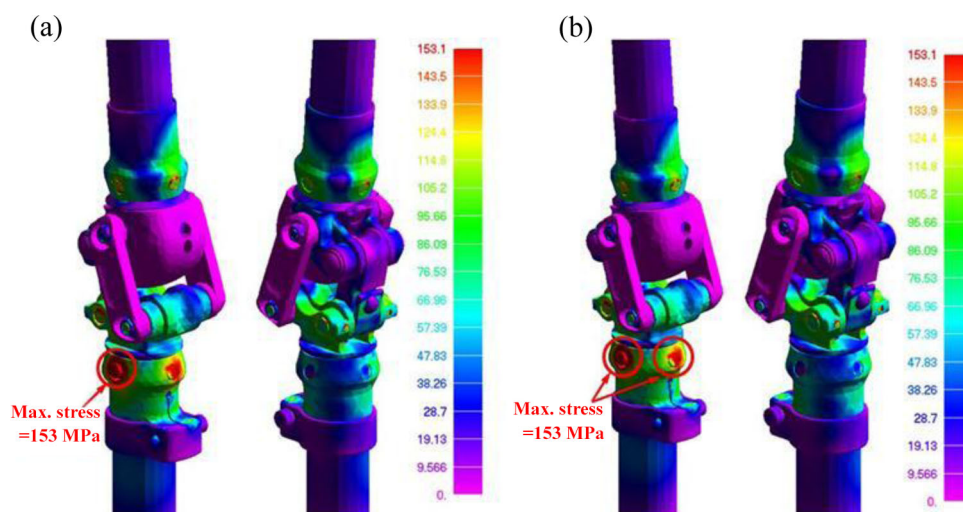


Figure 5. Stress distributions on a prosthetic knee under static strength test. (a) Loading condition I. (b) Loading condition II.

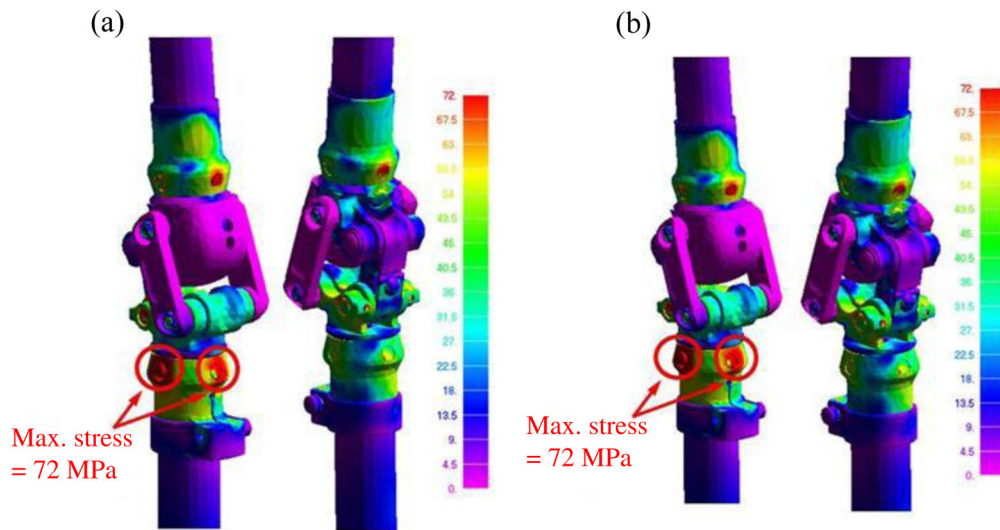


Figure 6. Stress distributions on a prosthetic knee under one loading cycle of cyclic strength test. (a) Loading condition I. (b) Loading condition II.

Additionally, the stress distributions estimated from one loading cycle of the cyclic strength test under loading conditions I and II are shown in Figure 6. These stress distributions are then given to the MSC Fatigue module to predict the fatigue life. Figure 7 reveals the fatigue

strength of the prosthesis in both loading conditions. The fringe contours on the side indicate the number of cycles that the prosthesis is able to withstand. It is clear from the figure that this prosthetic knee may be able to exceed the border line of 3×10^6 cycles specified in the standard.



Figure 7. Fatigue strength prediction of a prosthetic knee. (a) Loading condition I. (b) Loading condition II.

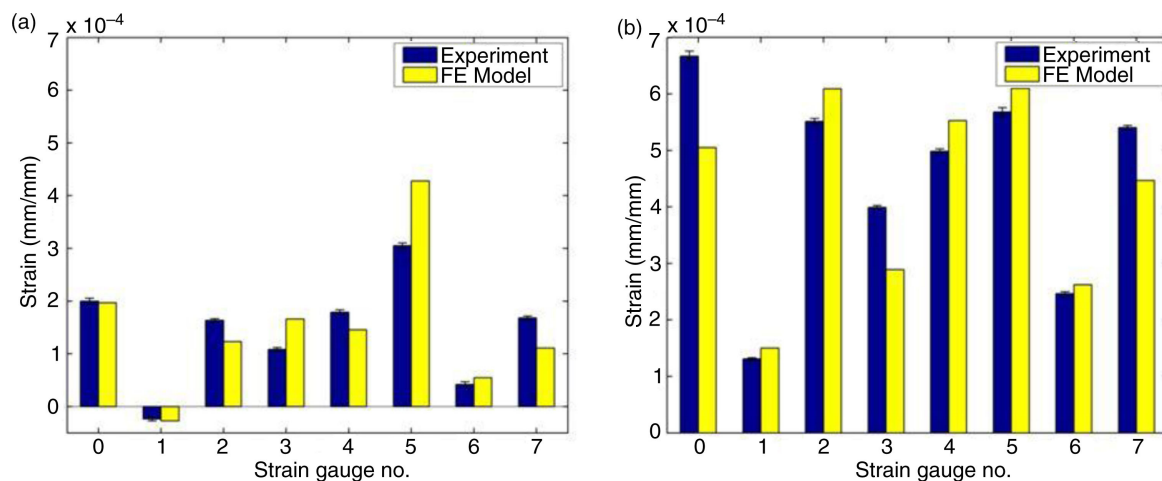


Figure 8. Comparisons of strains obtained from experiments and finite element model under static strength test. (a) Loading condition I. (b) Loading condition II.

Validation results

The structural strains obtained from the finite element model are validated with those measured from the experiments to determine the accuracy of model. Figure 8 is the comparison of strains acquired from the model and experiments in the loading conditions I and II of the static strength test. In these figures, the horizontal axis denotes the strain gauge identification number defined in Figure 3, and the vertical axis presents the strain values. As previously mentioned, five repetitions are performed for each loading condition; hence, the mean strains along with their standard deviations are plotted in the figure. Based on

these results, the means of absolute percentage error are figured to be 27 and 15% for the loading conditions I and II, respectively.

The strain comparisons in the case of the cyclic strength test are illustrated in Figure 9. As in the case of a static test, the strains indicated in the figure are the mean values estimated from five repetitions. From the figures, the averages of absolute percentage error for the loading conditions I and II are 25 and 22%, respectively. Based on these validation results in all cases, it is clear that most of the strains predicted by the model are fairly close to the ones measured from the physical tests. Besides, the physical

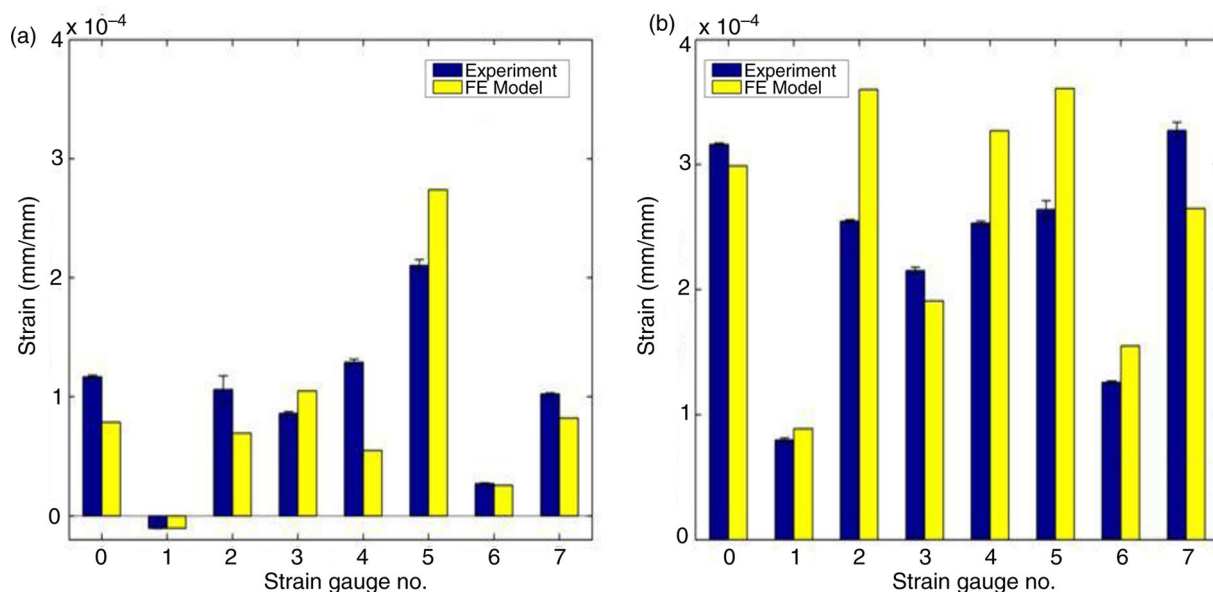


Figure 9. Comparisons of strains obtained from experiments and finite element model under cyclic strength test. (a) Loading condition I. (b) Loading condition II.

specimen of the prosthetic knee survives the physical cyclic strength test without any failures as predicted in the previous section, showing the fidelity of the fatigue life evaluation. Thus, it may be concluded that the finite element model implemented in this work is a potential candidate to evaluate the strength of prosthesis.

Conclusions

This work outlines the virtual static and cyclic strength tests of a four-bar linkage prosthetic knee. The test procedures are obtained from the ISO 10328:2006 standard. FEM is applied to construct a model of the prosthesis under the tests. The technique of nonlinear dynamic analysis is employed to estimate the stress distributions induced on the structure of prosthesis. The simulation results show that the prosthesis sample is sufficiently strong enough to surpass both of the static and cyclic strength tests. Moreover, the experimental validation is conducted to determine the accuracy of the stress prediction. The validation metric used in this work is a structural strain. Thus, to measure the physical strains, a physical prosthetic knee is instrumented with strain gauges and placed in a servo-pneumatic test machine. The prosthesis sample is subjected to five repetitions for each testing condition. The validation results, further, reveal that the proposed finite-element model is able to sufficiently compute the structural stresses in all cases. In the case of the static test, the mean absolute percentage errors are 27 and 15% for the loading conditions I and II, respectively. Besides, the analytical results deviate from the experimental ones by 25 and 22% for the loading conditions I and II of the cyclic test, respectively. The prosthesis specimen also survives the physical fatigue test without any failures as predicted in the simulations.

Conflicts of interest and funding

The authors have not received any funding or benefits from industry or elsewhere to conduct this study.

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